4.1 Exponential and Logarithmic Functions

INTRODUCTION The real exponential and logarithmic functions play an important role in the study of real analysis and differential equations. In this section we define and study the complex analogues of these functions. In the first part of this section, we study the complex exponential function e^z, which has already been introduced in Sections 1.6 and 2.1. One concept that has not been discussed previously but will be addressed in this section is the exponential mapping $w = e^z$. In the second half of this section, we introduce the complex logarithm $\ln z$ to solve exponential equations of the form $e^w = z$. If x is a fixed positive *real* number, then there is a *single* solution to the equation $e^y = x$, namely the value $y = \log_e x$. However, we will see that when z is a fixed nonzero *complex* number there are *infinitely* many solutions to the equation $e^w = z$. Therefore, the complex logarithm $\ln z$ is a "multiple-valued function" in the interpretation of this term given in Section 2.4. The principal value of the complex logarithm will be defined to be a (single-valued) function that assigns to the complex input z one of the multiple values of $\ln z$. This principal value function will be shown to be an inverse function of the exponential function e^z defined on a suitably restricted domain of the complex plane. We conclude the section with a discussion of the analyticity of branches of the logarithm.

4.1.1 Complex Exponential Function

Exponential Function and its Derivative We begin by repeating the definition of the complex exponential function given in Section 2.1.

DEFINITION 4.1.1 Complex Exponential Function

The function e^z defined by

$$e^z = e^x \cos y + i e^x \sin y \tag{1}$$

is called the **complex exponential function**.

A review of the real exponential and logarithmic functions can be found in Section 4.1 of the student study guide. ▶

One reason why it is natural to call this function the *exponential* function was pointed out in Section 2.1. Namely, the function defined by (1) agrees with the real exponential function when z is real. That is, if z is real, then $z = x + 0i$, and Definition 4.1.1 gives:

$$e^{x+0i} = e^x (\cos 0 + i \sin 0) = e^x (1 + i \cdot 0) = e^x. \tag{2}$$

The complex exponential function also shares important differential properties of the real exponential function. Recall that two important properties of the real exponential function are that e^x is differentiable everywhere and that $\dfrac{d}{dx} e^x = e^x$ for all x. The complex exponential function e^z has similar properties.

THEOREM 4.1.1 Analyticity of e^z

The exponential function e^z is entire and its derivative is given by:

$$\frac{d}{dz} e^z = e^z. \tag{3}$$

Proof In order to establish that e^z is entire, we use the criterion for analyticity given in Theorem 3.3.2. We first note that the real and imaginary parts, $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$, of e^z are continuous real functions and have continuous first-order partial derivatives for all (x, y). In addition, the Cauchy-Riemann equations in u and v are easily verified:

$$\frac{\partial u}{\partial x} = e^x \cos y = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x}.$$

Therefore, the exponential function e^z is entire by Theorem 3.3.2. By (9) of Section 3.3, the derivative of e^z is:

$$\frac{d}{dz}e^z = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^z. \qquad \blacksquare$$

Using the fact that the real and imaginary parts of an analytic function are harmonic conjugates, we can also show that the *only* entire function f that agrees with the real exponential function e^x for real input and that satisfies the differential equation $f'(z) = f(z)$ is the complex exponential function e^z defined by (1). See Problem 50 in Exercises 4.1.

EXAMPLE 1 Derivatives of Exponential Functions

Find the derivative of each of the following functions:

(**a**) $iz^4\left(z^2 - e^z\right)$ and (**b**) $e^{z^2-(1+i)z+3}$.

Solution (**a**) Using (3) and the product rule (4) in Section 3.2:

$$\begin{aligned}\frac{d}{dz}\left[iz^4\left(z^2 - e^z\right)\right] &= iz^4\left(2z - e^z\right) + 4iz^3\left(z^2 - e^z\right)\\ &= 6iz^5 - iz^4e^z - 4iz^3e^z.\end{aligned}$$

(**b**) Using (3) and the chain rule (6) in Section 3.2:

$$\frac{d}{dz}\left[e^{z^2-(1+i)z+3}\right] = e^{z^2-(1+i)z+3} \cdot (2z - 1 - i). \qquad \blacksquare$$

Modulus, Argument, and Conjugate The modulus, argument, and conjugate of the exponential function are easily determined from (1). If we express the complex number $w = e^z$ in polar form:

$$w = e^x \cos y + ie^x \sin y = r(\cos\theta + i\sin\theta),$$

then we see that $r = e^x$ and $\theta = y + 2n\pi$, for $n = 0, \pm1, \pm2, \ldots$. Because r is the modulus and θ is an argument of w, we have:

$$|e^z| = e^x \tag{4}$$

and

$$\arg(e^z) = y + 2n\pi, \quad n = 0, \pm1, \pm2, \ldots. \tag{5}$$

We know from calculus that $e^x > 0$ for all real x, and so it follows from (4) that $|e^z| > 0$. This implies that $e^z \neq 0$ for all complex z. Put another way, the point $w = 0$ is not in the range of the complex function $w = e^z$. Equation (4) does not, however, rule out the possibility that e^z is a negative real number. In fact, you should verify that if, say, $z = \pi i$, then $e^{\pi i}$ is real and $e^{\pi i} < 0$.

A formula for the conjugate of the complex exponential e^z is found using properties of the real cosine and sine functions. Since the real cosine function is even,

we have $\cos y = \cos(-y)$ for all y, and since the real sine function is odd, we have $-\sin y = \sin(-y)$ for all y, and so:

$$\overline{e^z} = e^x \cos y - ie^x \sin y = e^x \cos(-y) + ie^x \sin(-y) = e^{x-iy} = e^{\bar{z}}.$$

Therefore, for all complex z, we have shown:

$$\overline{e^z} = e^{\bar{z}}. \tag{6}$$

Algebraic Properties In Theorem 4.1.1 we proved that differentiating the complex exponential function is, in essence, the same as differentiating the real exponential function. These two functions also share the following algebraic properties.

THEOREM 4.1.2 Algebraic Properties of e^z

If z_1 and z_2 are complex numbers, then

(*i*) $e^0 = 1$

(*ii*) $e^{z_1}e^{z_2} = e^{z_1+z_2}$

(*iii*) $\dfrac{e^{z_1}}{e^{z_2}} = e^{z_1-z_2}$

(*iv*) $(e^{z_1})^n = e^{nz_1}$, $n = 0, \pm 1, \pm 2, \ldots$.

Proof of (*i*) and (*ii*) (*i*) The proof of property (*i*) follows from the observation that the complex exponential function agrees with the real exponential function for real input. That is, by (2), we have $e^{0+0i} = e^0$, and we know that, for the real exponential function, $e^0 = 1$.

(*ii*) Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. By Definition 4.1.1, we have:

$$\begin{aligned} e^{z_1}e^{z_2} &= \left(e^{x_1}\cos y_1 + ie^{x_1}\sin y_1\right)\left(e^{x_2}\cos y_2 + ie^{x_2}\sin y_2\right) \\ &= e^{x_1+x_2}(\cos y_1 \cos y_2 - \sin y_1 \sin y_2) \\ &\quad + ie^{x_1+x_2}(\sin y_1 \cos y_2 + \cos y_1 \sin y_2). \end{aligned}$$

Using the addition formulas for the real cosine and sine functions given on page 18 in Section 1.3, we can rewrite the foregoing expression as:

$$e^{z_1}e^{z_2} = e^{x_1+x_2}\cos(y_1 + y_2) + ie^{x_1+x_2}\sin(y_1 + y_2). \tag{7}$$

From (1), the right-hand side of (7) is $e^{z_1+z_2}$. Therefore, we have shown that $e^{z_1}e^{z_2} = e^{z_1+z_2}$. ☰

The proofs of Theorem 4.1.2(*iii*) and 4.1.2(*iv*) follow in a similar manner. See Problems 47 and 48 in Exercises 4.1.

Periodicity The most striking difference between the real and complex exponential functions is the periodicity of e^z. We say that a complex function f is **periodic** with period T if $f(z + T) = f(z)$ for all complex z. The real exponential function is not periodic, but the complex exponential function is because it is defined using the real cosine and sine functions, which are periodic. In particular, by (1) and Theorem 4.1.2(*ii*) we have $e^{z+2\pi i} = e^z e^{2\pi i} = e^z(\cos 2\pi + i \sin 2\pi)$. Since $\cos 2\pi = 1$ and $\sin 2\pi = 0$, this simplifies to:

$$e^{z+2\pi i} = e^z. \tag{8}$$

In summary, we have shown that:

- *The complex exponential function e^z is periodic with a pure imaginary period $2\pi i$.*

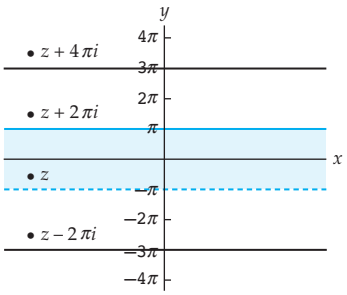


FIGURE 4.1.1 The fundamental region of e^z

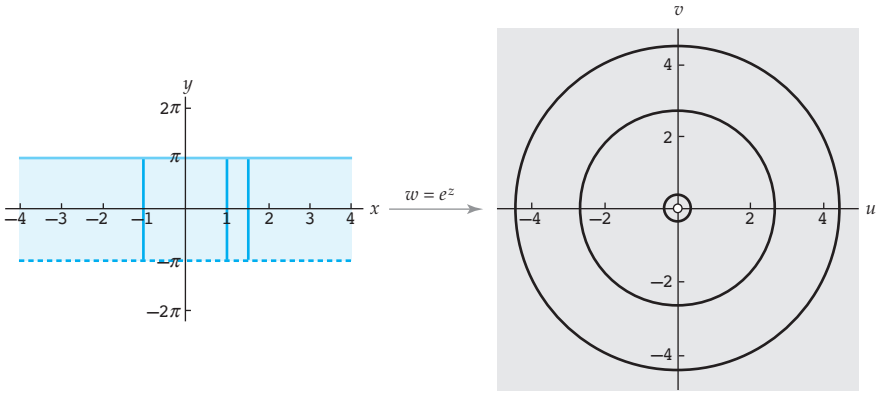
Because (8) holds for all values of z we also have $e^{(z+2\pi i)+2\pi i} = e^{z+2\pi i}$. This fact combined with (8) implies that $e^{z+4\pi i} = e^z$. Now by repeating this process we find that $e^{z+2n\pi i} = e^z$ for $n = 0, \pm 1, \pm 2, \ldots$. Thus, the complex exponential function is not one-to-one, and all values e^z are assumed in any infinite horizontal strip of width 2π in the z-plane. For example, all values of the function e^z are assumed in the set $-\infty < x < \infty$, $y_0 < y \leq y_0 + 2\pi$, where y_0 is a real constant. In **FIGURE 4.1.1** we divide the complex plane into horizontal strips obtained by setting y_0 equal to any odd multiple of π. If the point z is in the infinite horizontal strip $-\infty < x < \infty$, $-\pi < y \leq \pi$, shown in color in Figure 4.1.1, then the values $f(z) = e^z$, $f(z + 2\pi i) = e^{z+2\pi i}$, $f(z - 2\pi i) = e^{z-2\pi i}$, and so on are the same. The infinite horizontal strip defined by:

$$-\infty < x < \infty, \quad -\pi < y \leq \pi, \tag{9}$$

is called the **fundamental region** of the complex exponential function.

The Exponential Mapping Since all values of the complex exponential function are assumed in the fundamental region defined by (9), the image of this region is the same as the image of the entire complex plane. In order to determine the image of the fundamental region under $w = e^z$, we note that this region consists of the collection of vertical line segments $z(t) = a + it$, $-\pi < t \leq \pi$, where a is any real number. By (11) of Section 2.2, the image of the line segment $z(t) = a + it$, $-\pi < t \leq \pi$, under $w = e^z$ is parametrized by $w(t) = e^{z(t)} = e^{a+it} = e^a e^{it}$, $-\pi < t \leq \pi$. Now from (10) of Section 2.2, this image is a circle centered at the origin with radius e^a. Because a can be any real number, the radius e^a of this circle can be any nonzero positive real number. Thus, the image of the fundamental region under the exponential mapping consists of the collection of all circles centered at the origin with nonzero radius. In other words, the image of the fundamental region (9) under $w = e^z$ is the set of all complex w with $w \neq 0$, or, equivalently, the set $|w| > 0$.*

The mapping $w = e^z$ of the fundamental region is shown in **FIGURE 4.1.2**. Each vertical line segment shown in color in Figure 4.1.2(a) is mapped onto a circle



(a) Vertical line segments in the fundamental region

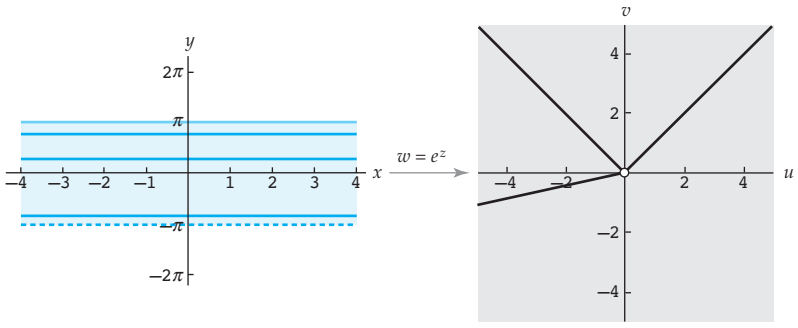
(b) Images of the line segments in (a) are circles

FIGURE 4.1.2 The image of the fundamental region under $w = e^z$

*This set is sometimes called the punctured complex plane.

shown in black in Figure 4.1.2(b). As the x-intercept of a vertical line segment increases, the radius of its image increases. Therefore, the leftmost line segment maps onto the innermost circle, the middle line segment maps onto the middle circle, and the rightmost line segment maps onto the outermost circle.

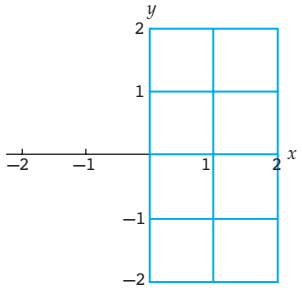
There was nothing particularly special about using vertical line segments to determine the image of the fundamental region. The image can also be found using horizontal lines. If the horizontal line $y = b$ is parametrized by $z(t) = t + ib$, $-\infty < t < \infty$, then its image under $w = e^z$ is given by $w(t) = e^t e^{ib}$, $-\infty < t < \infty$. After changing to the parameter $s = e^t$, the image is given by $W(s) = e^{ib}s$, $0 < s < \infty$. By (8) of Section 2.2, this is the set of points $w \neq 0$ in the ray emanating from the origin and containing the point $e^{ib} = \cos b + i \sin b$. That is, the image of the horizontal line $y = b$ under the mapping $w = e^z$ is the ray $\arg(w) = b$. We represent this property of the complex exponential mapping in FIGURE 4.1.3. Each horizontal line shown in color in Figure 4.1.3(a) is mapped onto a ray shown in black in Figure 4.1.3(b). As the y-intercept of a horizontal line increases, the angle the image ray makes with the positive u-axis increases. Therefore, the bottom-most line maps onto the ray in the third quadrant, the middle line maps onto the ray in the first quadrant, and the top-most line maps onto the ray in the second quadrant. The following summarizes properties of the mapping $w = e^z$.



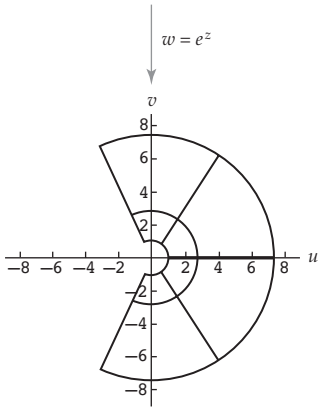
(a) Horizontal lines in the fundamental region

(b) Images of the lines in (a)

FIGURE 4.1.3 The mapping $w = e^z$



(a) Figure for Example 2



(b) Image of the grid in (a)

FIGURE 4.1.4 The mapping $w = e^z$

Exponential Mapping Properties

(*i*) $w = e^z$ *maps the fundamental region* $-\infty < x < \infty$, $-\pi < y \leq \pi$, *onto the set* $|w| > 0$.

(*ii*) $w = e^z$ *maps the vertical line segment* $x = a$, $y_0 \leq y \leq y_1$, *onto the circular arc* $|w| = e^a$, $y_0 \leq \arg(w) \leq y_1$.

(*iii*) $w = e^z$ *maps the horizontal line segment* $y = b$, $x_0 < x < x_1$, *onto the segment of the ray* $\arg(w) = b$, $e^{x_0} \leq |w| \leq e^{x_1}$.

EXAMPLE 2 Exponential Mapping of a Grid

Find the image of the grid shown in FIGURE 4.1.4(a) under $w = e^z$.

Solution Using property (*ii*) of the exponential mapping, we have that the images of the vertical line segments $x = 0$, 1, and 2, $-2 \leq y \leq 2$, are the circular arcs $|w| = e^0$, e^1, and e^2, $-2 \leq \arg(w) \leq 2$, respectively. On the other hand, by property (*iii*), the

images of the horizontal line segments $y = -2, -1, 0, 1$, and 2, $0 \le x \le 2$, are the segments of the rays $\arg(w) = -2, -1, 0, 1$, and 2, $1 \le |w| \le e^2$, respectively. Therefore, the vertical line segments shown in color in Figure 4.1.4(a) map onto the circular arcs shown in black in Figure 4.1.4(b) with the line segment $x = a$ mapping onto the arc with radius e^a. In addition, the horizontal line segments shown in color in Figure 4.1.4(a) map onto the black line segments in Figure 4.1.4(b) with the line segment $y = b$ mapping onto the line segment making an angle of b radians with the positive u-axis.

4.1.2 Complex Logarithmic Function

In real analysis, the natural logarithm function $\ln x$ is often defined as an inverse function of the real exponential function e^x. From this point on, we will use the alternative notation $\log_e x$ to represent the real exponential function. Because the real exponential function is one-to-one on its domain $\mathbf{R}$, there is no ambiguity involved in defining this inverse function. The situation is very different in complex analysis because the complex exponential function e^z is not a one-to-one function on its domain $\mathbf{C}$. In fact, given a fixed nonzero complex number z, the equation $e^w = z$ has infinitely many solutions. In order to see that this is so, assume that $w = u + iv$. If $e^w = z$, then $|e^w| = |z|$ and $\arg(e^w) = \arg(z)$. From (4) and (5), it follows that $e^u = |z|$ and $v = \arg(z)$, or, equivalently, $u = \log_e |z|$ and $v = \arg(z)$. Therefore, given a nonzero complex number z we have shown that:

Note: $\log_e x$ will be used to denote the real logarithm. ▶

$$\text{If } e^w = z, \text{ then } w = \log_e |z| + i \arg(z). \tag{10}$$

Because there are infinitely many arguments of z, (10) gives infinitely many solutions w to the equation $e^w = z$. The set of values given by (10) defines a multiple-valued function $w = G(z)$, as described in Section 2.4, which is called the complex logarithm of z and denoted by $\ln z$. The following definition summarizes this discussion.

DEFINITION 4.1.2 Complex Logarithm

The multiple-valued function $\ln z$ defined by:

$$\ln z = \log_e |z| + i \arg(z) \tag{11}$$

is called the **complex logarithm**.

Note: $\ln z$ will be used to denote the complex logarithm. ▶

Hereafter, the notation $\ln z$ will always be used to denote the multiple-valued *complex* logarithm. By switching to exponential notation $z = re^{i\theta}$ in (11), we obtain the following alternative description of the complex logarithm:

$$\ln z = \log_e r + i(\theta + 2n\pi), \quad n = 0, \pm 1, \pm 2, \ldots . \tag{12}$$

From (10) we see that the complex logarithm can be used to find all solutions to the exponential equation $e^w = z$ when z is a nonzero complex number.

EXAMPLE 3 Solving Exponential Equations

Find all complex solutions to each of the following equations.

(a) $e^w = i$ **(b)** $e^w = 1 + i$ **(c)** $e^w = -2$

Solution For each equation $e^w = z$, the set of solutions is given by $w = \ln z$ where $\ln z$ is found using Definition 4.1.2.

(**a**) For $z = i$, we have $|z| = 1$ and $\arg(z) = \pi/2 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln i = \log_e 1 + i\left(\frac{\pi}{2} + 2n\pi\right).$$

Since $\log_e 1 = 0$, this simplifies to:

$$w = \frac{(4n+1)\pi}{2}i, \quad n = 0, \pm 1, \pm 2, \ldots.$$

Therefore, each of the values:

$$w = \ldots, -\frac{7\pi}{2}i, -\frac{3\pi}{2}i, \frac{\pi}{2}i, \frac{5\pi}{2}i, \ldots$$

satisfies the equation $e^w = i$.

(**b**) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\arg(z) = \pi/4 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln(1+i) = \log_e \sqrt{2} + i\left(\frac{\pi}{4} + 2n\pi\right).$$

Because $\log_e \sqrt{2} = \frac{1}{2}\log_e 2$, this can be rewritten as:

$$w = \frac{1}{2}\log_e 2 + \frac{(8n+1)\pi}{4}i, \quad n = 0, \pm 1, \pm 2, \ldots.$$

Each value of w is a solution to $e^w = 1 + i$.

(**c**) Again we use (11). Since $z = -2$, we have $|z| = 2$ and $\arg(z) = \pi + 2n\pi$, and so:

$$w = \ln(-2) = \log_e 2 + i(\pi + 2n\pi).$$

That is,

$$w = \log_e 2 + (2n+1)\pi i, \quad n = 0, \pm 1, \pm 2, \ldots.$$

Each value of w satisfies the equation $e^w = -2$. ☰

Logarithmic Identities Definition 4.1.2 can be used to prove that the complex logarithm satisfies the following identities, which are analogous to identities for the real logarithm.

THEOREM 4.1.3 Algebraic Properties of ln *z*

If z_1 and z_2 are nonzero complex numbers and n is an integer, then

(*i*) $\ln(z_1 z_2) = \ln z_1 + \ln z_2$

(*ii*) $\ln\left(\frac{z_1}{z_2}\right) = \ln z_1 - \ln z_2$

(*iii*) $\ln z_1^n = n \ln z_1$.

Proof of (*i*) By Definition 4.1.2,

$$\begin{aligned}\ln z_1 + \ln z_2 &= \log_e |z_1| + i \arg(z_1) + \log_e |z_2| + i \arg(z_2) \\ &= \log_e |z_1| + \log_e |z_2| + i (\arg(z_1) + \arg(z_2)) . \end{aligned} \tag{13}$$

Because the real logarithm has the property $\log_e a + \log_e b = \log_e (ab)$ for $a > 0$ and $b > 0$, we can write $\log_e |z_1 z_2| = \log_e |z_1| + \log_e |z_2|$. Moreover, from (8) of Section 1.3, we have $\arg(z_1) + \arg(z_2) = \arg(z_1 z_2)$. Therefore, (13) can be rewritten as:

$$\ln z_1 + \ln z_2 = \log_e |z_1 z_2| + i \arg(z_1 z_2) = \ln(z_1 z_2) .$$

Proofs of Theorems 4.1.3(*ii*) and 4.1.3(*iii*) are similar. See Problems 53 and 54 in Exercises 4.1.

Principal Value of a Complex Logarithm It is interesting to note that the *complex* logarithm of a positive real number has infinitely many values. For example, the *complex* logarithm $\ln 5$ is the set of values $1.6094 + 2n\pi i$, where n is any integer, whereas the *real* logarithm $\log_e 5$ has a single value: $\log_e 5 \approx 1.6094$. The unique value of $\ln 5$ corresponding to $n = 0$ is the same as the value of the real logarithm $\log_e 5$. In general, this value of the complex logarithm is called the **principal value of the complex logarithm** since it is found by using the principal argument $\text{Arg}(z)$ in place of the argument $\arg(z)$ in (11). We denote the principal value of the logarithm by the symbol $\text{Ln}\, z$. Thus, the expression $f(z) = \text{Ln}\, z$ defines a function, whereas, $F(z) = \ln z$ defines a multiple-valued function. We summarize this discussion in the following definition.

Notation used throughout this text ▶

DEFINITION 4.1.3 Principal Value of the Complex Logarithm

The complex function $\text{Ln}\, z$ defined by:

$$\text{Ln}\, z = \log_e |z| + i \text{Arg}(z) \tag{14}$$

is called the **principal value of the complex logarithm**.

We will use the terms *logarithmic function* and *logarithm* to refer to both the multiple-valued function $\ln z$ and the function $\text{Ln}\, z$. In context, however, it should be clear to which of these we are referring. From (14), we see that the principal value of the complex logarithm can also be given by:

$$\text{Ln}\, z = \log_e r + i\theta, \quad -\pi < \theta \leq \pi . \tag{15}$$

EXAMPLE 4 Principal Value of the Complex Logarithm

Compute the principal value of the complex logarithm $\text{Ln}\, z$ for

(a) $z = i$ **(b)** $z = 1 + i$ **(c)** $z = -2$

Solution In each part we apply (14) of Definition 4.1.3.

(a) For $z = i$, we have $|z| = 1$ and $\text{Arg}(z) = \pi/2$, and so:

$$\text{Ln}\, i = \log_e 1 + \frac{\pi}{2} i .$$

However, since $\log_e 1 = 0$, this simplifies to:

$$\text{Ln}\, i = \frac{\pi}{2} i .$$

(**b**) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\text{Arg}(z) = \pi/4$, and so:

$$\text{Ln}(1 + i) = \log_e \sqrt{2} + \frac{\pi}{4} i.$$

Because $\log_e \sqrt{2} = \frac{1}{2} \log_e 2$, this can also be written as:

$$\text{Ln}(1 + i) = \frac{1}{2} \log_e 2 + \frac{\pi}{4} i \approx 0.3466 + 0.7854i.$$

(**c**) For $z = -2$, we have $|z| = 2$ and $\text{Arg}(z) = \pi$, and so:

$$\text{Ln}(-2) = \log_e 2 + \pi i \approx 0.6931 + 3.1416i.$$

Observe that each of the values found in parts (a)–(c) could have also been found by setting $n = 0$ in the expressions for $\ln z$ from Example 3. ☰

Note ▶ It is important to note that the identities for the complex logarithm in Theorem 4.1.3 are *not* necessarily satisfied by the principal value of the complex logarithm. For example, it is *not* true that $\text{Ln}(z_1 z_2) = \text{Ln}\, z_1 + \text{Ln}\, z_2$ for all complex numbers z_1 and z_2 (although it may be true for *some* complex numbers). See Problem 55 in Exercises 4.1.

Ln *z* as an Inverse Function Because $\text{Ln}\, z$ is *one* of the values of the complex logarithm $\ln z$, it follows from (10) that:

$$e^{\text{Ln}z} = z \text{ for all } z \neq 0. \tag{16}$$

This suggests that the logarithmic function $\text{Ln}\, z$ is an inverse function of exponential function e^z. Because the complex exponential function is not one-to-one on its domain, this statement is not completely accurate. Rather, the relationship between these functions is similar to the relationship between the squaring function z^2 and the principal square root function $z^{1/2} = \sqrt{|z|} e^{i\,\text{Arg}(z)/2}$ defined by (7) in Section 2.4. The exponential function must first be restricted to a domain on which it is one-to-one in order to have a well-defined inverse function. In Problem 52 in Exercises 4.1, you will be asked to show that e^z is a one-to-one function on the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, shown in Figure 4.1.1.

We now show that if the domain of e^z is restricted to the fundamental region, then the principal value of the complex logarithm $\text{Ln}\, z$ is its inverse function. To justify this claim, consider a point $z = x + iy$ in the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$. From (4) and (5), we have that $|e^z| = e^x$ and $\arg(e^z) = y + 2n\pi$, n an integer. Thus, y is an argument of e^z. Since z is in the fundamental region, we also have $-\pi < y \leq \pi$, and from this it follows that y is the principal argument of e^z. That is, $\text{Arg}(e^z) = y$. In addition, for the real logarithm we have $\log_e e^x = x$, and so from Definition 4.1.3 we obtain:

$$\begin{aligned} \text{Ln}\, e^z &= \log_e |e^z| + i\,\text{Arg}\,(e^z) \\ &= \log_e e^x + iy \\ &= x + iy. \end{aligned}$$

Thus, we have shown that:

$$\text{Ln}\, e^z = z \text{ if } -\infty < x < \infty \quad \text{and} \quad -\pi < y \leq \pi. \tag{17}$$

From (16) and (17), we conclude that $\text{Ln}\, z$ is the inverse function of e^z defined on the fundamental region. The following summarizes the relationship between these functions.

Ln z as an Inverse Function of e^z

If the complex exponential function $f(z) = e^z$ is defined on the fundamental region $-\infty < x < \infty, -\pi < y \leq \pi$, then f is one-to-one and the inverse function of f is the principal value of the complex logarithm $f^{-1}(z) = \operatorname{Ln} z$.

Bear in mind that (16) holds for all nonzero complex numbers z, but (17) only holds if z is in the fundamental region. For example, for the point $z = 1 + \frac{3}{2}\pi i$, which is not in the fundamental region, we have:

$$\operatorname{Ln} e^{1+3\pi i/2} = 1 - \frac{1}{2}\pi i \neq 1 + \frac{3}{2}\pi i.$$

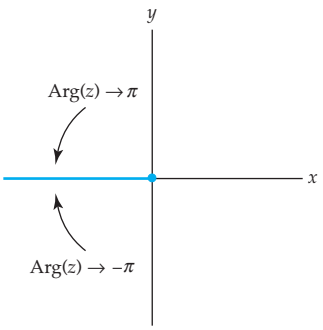


FIGURE 4.1.5 Ln z is discontinuous at $z = 0$ and on the negative real axis.

Analyticity The principal value of the complex logarithm Ln z is discontinuous at the point $z = 0$ since this function is not defined there. This function also turns out to be discontinuous at every point on the negative real axis. This is intuitively clear since the value of Ln z a point z near the negative x-axis in the second quadrant has imaginary part close to π, whereas the value of a nearby point in the third quadrant has imaginary part close to $-\pi$. See FIGURE 4.1.5. The function Ln z is, however, continuous on the set consisting of the complex plane excluding the nonpositive real axis. To see that this is so, we appeal to Theorem 3.1.3, which states that a complex function $f(z) = u(x, y) + iv(x, y)$ is continuous at a point $z = x + iy$ if and only if both u and v are continuous real functions at (x, y). By (14), the real and imaginary parts of Ln z are $u(x, y) = \log_e |z| = \log_e \sqrt{x^2 + y^2}$ and $v(x, y) = \operatorname{Arg}(z)$, respectively. From multivariable calculus we have that the function $u(x, y) = \log_e \sqrt{x^2 + y^2}$ is continuous at all points in the plane except $(0, 0)$ and from Problem 55 in Exercises 3.1 we have that the function $v(x, y) = \operatorname{Arg}(z)$ is continuous on the domain $|z| > 0, -\pi < \arg(z) < \pi$. Therefore, by Theorem 3.1.3, it follows that Ln z is a continuous function on the domain

$$|z| > 0, \quad -\pi < \arg(z) < \pi, \tag{18}$$

shown in gray in FIGURE 4.1.6. Put another way, the function f_1 defined by:

$$f_1(z) = \log_e r + i\theta, -\pi < \theta < \pi \tag{19}$$

is continuous on the domain in (18).

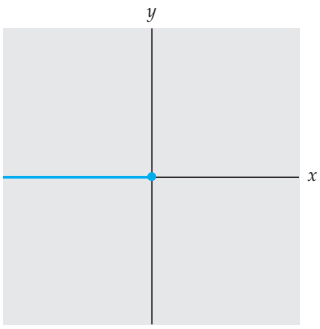


FIGURE 4.1.6 Branch cut for f_1

Since the function f_1 agrees with the principal value of the complex logarithm Ln z where they are both defined, it follows that f_1 assigns to the input z one of the values of the multiple-valued function $F(z) = \ln z$. Using the terminology of Section 3.1, we have shown that the function f_1 defined by (19) is a branch of the multiple-valued function $F(z) = \ln z$. (Recall that branches of a multiple-valued function F are denoted by f_1, f_2, and so on.) This branch is called the **principal branch of the complex logarithm**. The nonpositive real axis shown in color in FIGURE 4.1.6 is a **branch cut** for f_1 and the point $z = 0$ is a **branch point**. As the following theorem demonstrates, the branch f_1 is an analytic function on its domain.

THEOREM 4.1.4 Analyticity of the Principal Branch of ln z

The principal branch f_1 of the complex logarithm defined by (19) is an analytic function and its derivative is given by:

$$f_1'(z) = \frac{1}{z}. \tag{20}$$

Proof We prove that f_1 is analytic by using the polar coordinate analogue to Theorem 3.3.2 of Section 3.3. Because f_1 is defined on the domain given in (18), if z is a point in this domain, then we can write $z = re^{i\theta}$ with $-\pi < \theta < \pi$. Since the real and imaginary parts of f_1 are $u(r, \theta) = \log_e r$ and $v(r, \theta) = \theta$, respectively, we find that:

$$\frac{\partial u}{\partial r} = \frac{1}{r}, \qquad \frac{\partial v}{\partial \theta} = 1,$$
$$\frac{\partial v}{\partial r} = 0, \quad \text{and} \quad \frac{\partial u}{\partial \theta} = 0.$$

Thus, u and v satisfy the Cauchy-Riemann equations in polar coordinates (10) in Section 3.3:

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

Because u, v, and the first partial derivatives of u and v are continuous at all points in the domain given in (18), it follows from Theorem 3.3.2 that f_1 is analytic in this domain. In addition, from (11) of Section 3.3, the derivative of f_1 is given by:

$$f_1'(z) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = \frac{1}{re^{i\theta}} = \frac{1}{z}.$$

☰

Because $f_1(z) = \operatorname{Ln} z$ for each point z in the domain given in (18), it follows from Theorem 4.1.4 that $\operatorname{Ln} z$ is differentiable in this domain, and that its derivative is given by f_1'. That is, if $|z| > 0$ and $-\pi < \arg(z) < \pi$ then:

$$\frac{d}{dz}\operatorname{Ln} z = \frac{1}{z}. \tag{21}$$

EXAMPLE 5 Derivatives of Logarithmic Functions

Find the derivatives of the following functions in an appropriate domain:

(**a**) $z\operatorname{Ln} z$ and (**b**) $\operatorname{Ln}(z+1)$.

Solution (**a**) From the differentiation rules of Section 3.2 we have that the function $z\operatorname{Ln} z$ is differentiable at all points where both of the functions z and $\operatorname{Ln} z$ are differentiable. Because z is entire and $\operatorname{Ln} z$ is differentiable on the domain given in (18), it follows that $z\operatorname{Ln} z$ is differentiable on the domain defined by $|z| > 0$, $-\pi < \arg(z) < \pi$. In this domain the derivative is given by the product rule (4) of Section 3.2 and (21):

$$\frac{d}{dz}[z\operatorname{Ln} z] = z \cdot \frac{1}{z} + 1 \cdot \operatorname{Ln} z = 1 + \operatorname{Ln} z.$$

(**b**) The function $\operatorname{Ln}(z+1)$ is a composition of the functions $\operatorname{Ln} z$ and $z+1$. Because the function $z+1$ is entire, it follows from the chain rule that $\operatorname{Ln}(z+1)$ is differentiable at all points $w = z+1$ such that $|w| > 0$ and $-\pi < \arg(w) < \pi$. In other words, this function is differentiable at the point w whenever w does not lie on the nonpositive real axis. To determine the corresponding values of z for which $\operatorname{Ln}(z+1)$ is not differentiable, we first solve for z in terms of w to obtain $z = w - 1$. The equation $z = w - 1$ defines a linear mapping of the w-plane onto the z-plane given by translation by -1. Under this mapping the nonpositive real axis is mapped onto the

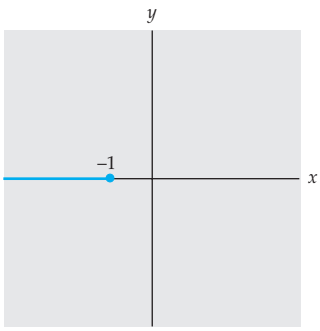


FIGURE 4.1.7 $\text{Ln}(z+1)$ is not differentiable on the ray shown in color.

ray emanating from $z = -1$ and containing the point $z = -2$ shown in color in FIGURE 4.1.7. Thus, if the point $w = z + 1$ is on the nonpositive real axis, then the point z is on the ray shown in Figure 4.1.7. This implies that $\text{Ln}(z+1)$ is differentiable at all points z that are not on this ray. For such points, the chain rule gives:

$$\frac{d}{dz}\text{Ln}(z+1) = \frac{1}{z+1} \cdot 1 = \frac{1}{z+1}.$$

Logarithmic Mapping

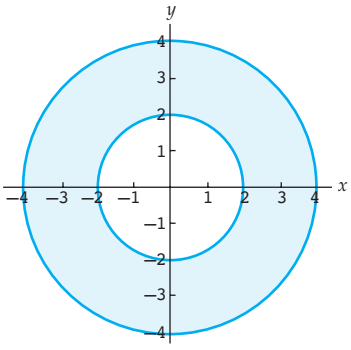
The complex logarithmic mapping $w = \text{Ln}\, z$ can be understood in terms of the exponential mapping $w = e^z$ since these functions are inverses of each other. For example, because $w = e^z$ maps the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, in the z-plane onto the set $|w| > 0$ in the w-plane, it follows that inverse mapping $w = \text{Ln}\, z$ maps the set $|z| > 0$ in the z-plane onto the region $-\infty < u < \infty$, $-\pi < v \leq \pi$, in the w-plane. Other properties of the exponential mapping can be similarly restated as properties of the logarithmic mapping. The following summarizes some of these properties.

Logarithmic Mapping Properties

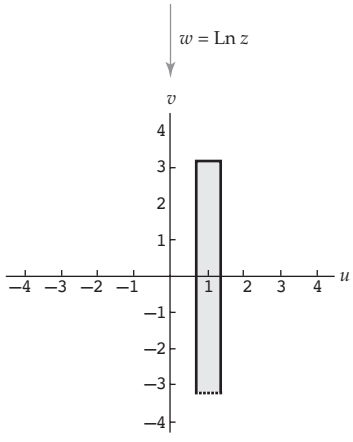
(*i*) $w = \text{Ln}\, z$ *maps the set* $|z| > 0$ *onto the region* $-\infty < u < \infty$, $-\pi < v \leq \pi$.

(*ii*) $w = \text{Ln}\, z$ *maps the circular arc* $|z| = r$, $\theta_0 \leq \text{Arg}(z) \leq \theta_1$, *onto the vertical line segment* $u = \log_e r$, $\theta_0 \leq v \leq \theta_1$.

(*iii*) $w = \text{Ln}\, z$ *maps the segment of the ray* $\text{Arg}(z) = \theta$, $r_0 \leq |z| \leq r_1$, *onto the horizontal line segment* $v = \theta$, $\log_e r_0 \leq u \leq \log_e r_1$.



(a) The annulus $2 \leq |z| \leq 4$



(b) The image of the annulus in (a)

FIGURE 4.1.8 The mapping $w = \text{Ln}\, z$

EXAMPLE 6 Logarithmic Mapping

Find the image of the annulus $2 \leq |z| \leq 4$ under the logarithmic mapping $w = \text{Ln}\, z$.

Solution From property (*ii*) of logarithmic mapping, the boundary circles $|z| = 2$ and $|z| = 4$ of the annulus map onto the vertical line segments $u = \log_e 2$ and $u = \log_e 4$, $-\pi < v \leq \pi$, respectively. In a similar manner, each circle $|z| = r$, $2 \leq r \leq 4$, maps onto a vertical line segment $u = \log_e r$, $-\pi < v \leq \pi$. Since the real logarithmic function is increasing on its domain, it follows that $u = \log_e r$ takes on all values in the interval $\log_e 2 \leq u \leq \log_e 4$ when $2 \leq r \leq 4$. Therefore, the image of the annulus $2 \leq |z| \leq 4$ shown in color in FIGURE 4.1.8(a) is the rectangular region $\log_e 2 \leq u \leq \log_e 4$, $-\pi < v \leq \pi$, shown in gray in Figure 4.1.8(b).

Other Branches of ln *z*

The principal branch of the complex logarithm f_1 defined in (19) is just one of many possible branches of the multiple-valued function $F(z) = \ln z$. We can define other branches of F by simply changing the interval defining θ in (18) to a different interval of length 2π. For example,

$$f_2(z) = \log_e r + i\theta, \quad -\frac{\pi}{2} < \theta < \frac{3\pi}{2},$$

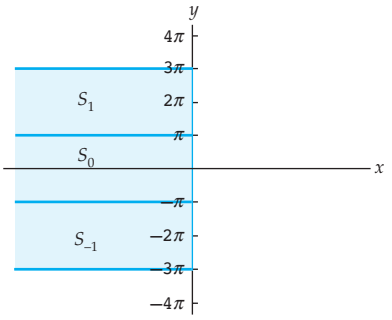
defines a branch of F whose branch cut is the nonpositive imaginary axis. You should verify that for the branch f_2 we have $f_2(1) = 0$, $f_2(2i) = \log_e 2 + \frac{1}{2}\pi i$, and $f_2(-1 - i) = \frac{1}{2}\log_e 2 + \frac{5}{4}\pi i$.

In the same way that we proved that the principal branch f_1 of the complex logarithm is analytic, we can also show that any branch

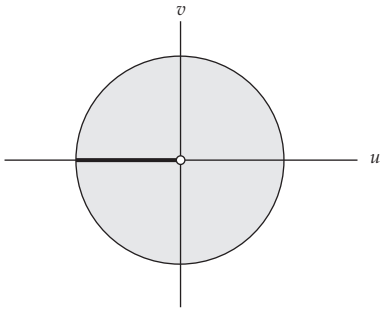
$$f_k(z) = \log_e r + i\theta, \quad \theta_0 < \theta < \theta_0 + 2\pi,$$

of $F(z) = \ln z$ is analytic on its domain, and its derivative is given by:

$$f_k'(z) = \frac{1}{z}.$$



(a) Collection of half-infinite strips S_n



(b) The image of each strip S_n is the punctured unit disk

FIGURE 4.1.9 The mapping $w = e^z$

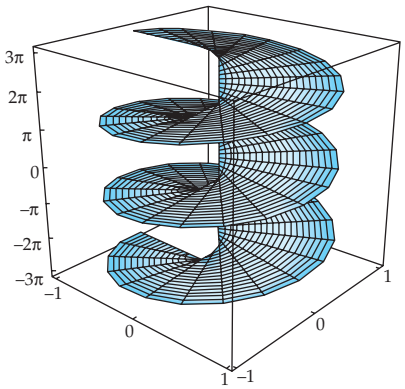


FIGURE 4.1.10 A Riemann surface for $w = e^z$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Although the complex exponential and logarithmic functions are similar to the real exponential and logarithmic functions in many ways, it is important to keep in mind their differences.

- The real exponential function is one-to-one, but the complex exponential is not.
- $\log_e x$ is a single-valued function, but $\ln z$ is multiple-valued.
- Many properties of real logarithms apply to the complex logarithm, such as $\ln(z_1 z_2) = \ln z_1 + \ln z_2$, but these properties don't always hold for the principal value $\operatorname{Ln} z$.

(*ii*) Since the complex exponential function is not one-to-one, we can use a Riemann surface, as described in the *Remarks* at the end of Section 2.4, to help visualize the mapping $w = e^z$. The Riemann surface that we construct will also help us to visualize the multiple-valued function $w = \ln z$. Consider the mapping $w = e^z$ on the half-plane $x \le 0$. Each half-infinite strip S_n defined by $(2n - 1)\pi < y \le (2n + 1)\pi$, $x \le 0$, for $n = 0, \pm 1, \pm 2, \ldots$ is mapped onto the punctured unit disk $0 < |w| \le 1$ shown in **FIGURE 4.1.9(b)** with the horizontal half-lines shown in color in Figure 4.1.9(a) mapping onto the segment $-1 \le u < 0$ shown in black in Figure 4.1.9(b). Thus, $w = e^z$ describes an infinite-to-one covering of the punctured unit disk. To visualize this covering, we imagine there being a different image disk B_n for each half-infinite strip S_n. Now cut each disk B_n open along the segment $-1 \le u < 0$. We construct a Riemann surface for $w = e^z$ by attaching, for each n, the cut disk B_n to the cut disk B_{n+1} along the edge that represents the image of the half-infinite line $y = (2n + 1)\pi$. We place this surface in xyz-space so that for each z in the half-plane, the images $\ldots\ z_{-1}, z_0, z_1, \ldots$ of z in $B_{n-1}, B_0, B_1, \ldots$, respectively, lie directly above the point $w = e^z$ in the xy-plane. See **FIGURE 4.1.10**. By projecting the points of the Riemann surface vertically down onto the xy-plane we see the infinite-to-one nature of the mapping $w = e^z$. Conversely, the multiple-valued function $F(z) = \ln z$ may be visualized by considering all points in the Riemann surface lying directly above a point in the xy-plane. These infinitely many points on the Riemann surface correspond to the infinitely many values of $F(z)$ in the half-plane $u \le 0$.

EXERCISES 4.1

Answers to selected odd-numbered problems begin on page ANS-12.

4.1.1 Complex Exponential Function

In Problems 1–4, find the derivative f' of the given function f.

1. $f(z) = z^2 e^{z+i}$

2. $f(z) = \dfrac{3e^{2z} - ie^{-z}}{z^3 - 1 + i}$

3. $f(z) = e^{iz} - e^{-iz}$

4. $f(z) = ie^{1/z}$

In Problems 5–8, write the given expression in terms of x and y.

5. $\left|e^{z^2 - z}\right|$

6. $\arg(e^{z - i/z})$

7. $\arg(e^{i(z+\bar{z})})$

8. $\overline{ie^z + 1}$

In Problems 9–12, express the given function f in the form $f(z) = u(x, y) + iv(x, y)$.

9. $f(z) = e^{-iz}$

10. $f(z) = e^{2\bar{z}+i}$

11. $f(z) = e^{z^2}$

12. $f(z) = e^{1/z}$

13. Use the sufficient conditions for differentiability from Section 3.3 to determine where the function $f(z) = e^{2\bar{z}+i}$ is differentiable.

14. Use the sufficient conditions for differentiability from Section 3.3 to determine where the function $f(z) = e^{z^2}$ is differentiable.

In Problems 15–20, find the image of the given set under the exponential mapping.

15. The line $y = -2$.

16. The line $x = 3$.

17. The infinite strip $1 < x \leq 2$.

18. The square with vertices at 0, 1, $1 + i$, and i.

19. The rectangle $0 \leq x \leq \log_e 2$, $-\pi/4 \leq y \leq \pi/2$.

20. The semi-infinite strip $-\infty < x \leq 0$, $0 \leq y \leq \pi$.

4.1.2 Complex Logarithmic Function

In Problems 21–26, find all complex values of the given logarithm.

21. $\ln(-5)$

22. $\ln(-ei)$

23. $\ln(-2 + 2i)$

24. $\ln(1 + i)$

25. $\ln\left(\sqrt{2} + \sqrt{6}i\right)$

26. $\ln\left(-\sqrt{3} + i\right)$

In Problems 27–32, write the principal value of the logarithm in the form $a + ib$.

27. $\operatorname{Ln}(6 - 6i)$

28. $\operatorname{Ln}\left(-e^2\right)$

29. $\operatorname{Ln}(-12 + 5i)$

30. $\operatorname{Ln}(3 - 4i)$

31. $\operatorname{Ln}\left[\left(1 + \sqrt{3}i\right)^5\right]$

32. $\operatorname{Ln}\left[(1 + i)^4\right]$

In Problems 33–36, find all complex values of z satisfying the given equation.

33. $e^z = 4i$

34. $e^{1/z} = -1$

35. $e^{z-1} = -ie^3$

36. $e^{2z} + e^z + 1 = 0$

In Problems 37–40, find a domain in which the given function f is differentiable; then find the derivative f'.

37. $f(z) = 3z^2 - e^{2iz} + i\operatorname{Ln} z$

38. $f(z) = (z+1)\operatorname{Ln} z$

39. $f(z) = \dfrac{\operatorname{Ln}(2z-i)}{z^2+1}$

40. $f(z) = \operatorname{Ln}\left(z^2+1\right)$

In Problems 41–46, find the image of the given set under the mapping $w = \operatorname{Ln} z$.

41. The ray $\arg(z) = \pi/6$.

42. The positive y-axis.

43. The circle $|z| = 4$.

44. The region in the first quadrant bounded by the circles $|z| = 1$ and $|z| = e$.

45. The annulus $3 \le |z| \le 5$.

46. The region outside the unit circle $|z| = 1$ and between the rays $\arg(z) = \pi/4$ and $\arg(z) = 3\pi/4$.

Focus on Concepts

47. Use (1) to prove that $e^{z_1}/e^{z_2} = e^{z_1 - z_2}$.

48. Use (1) and de Moivre's formula to prove that $(e^{z_1})^n = e^{nz_1}$, n an integer.

49. Determine where the complex function $e^{\bar{z}}$ is analytic.

50. In this problem, we will show that the complex exponential function defined by (1) is the only complex entire function f that agrees with the real exponential function e^x when z is real and that has the property $f'(z) = f(z)$ for all z.

(**a**) Assume that $f(z) = u(x,\ y) + iv(x,\ y)$ is an entire complex function for which $f'(z) = f(z)$. Explain why u and v satisfy the differential equations:

$$u_x(x,\ y) = u(x,\ y) \quad \text{and} \quad v_x(x,\ y) = v(x,\ y).$$

(**b**) Show that $u(x,\ y) = a(y)e^x$ and $v(x,\ y) = b(y)e^x$ are solutions to the differential equations in (a).

(**c**) Explain why the assumption that $f(z)$ agrees with the real exponential function for z real implies that $a(0) = 1$ and $b(0) = 0$.

(**d**) Explain why the functions $a(y)$ and $b(y)$ satisfy the system of differential equations:

$$a(y) - b'(y) = 0$$
$$a'(y) + b(y) = 0.$$

(**e**) Solve the system of differential equations in (d) subject to the initial conditions $a(0) = 1$ and $b(0) = 0$.

(**f**) Use parts (a)–(e) to show that the complex exponential function defined by (1) is the only complex entire function $f(z)$ that agrees with the real exponential function when z is real and that has the property $f'(z) = f(z)$ for all z.

51. Describe the image of the line $y = x$ under the exponential function. [*Hint*: Find a polar expression $r(\theta)$ of the image.]

52. Prove that e^z is a one-to-one function on the fundamental region $-\infty < x < \infty$, $-\pi < y \le \pi$.

53. Prove that $\ln\left(\dfrac{z_1}{z_2}\right) = \ln z_1 - \ln z_2$ for all nonzero complex numbers z_1 and z_2.

54. Prove that $\ln z_1^n = n \ln z_1$ for all nonzero complex numbers z_1 and all integers n.